

Data Analysis, SQL Cleaning, Missing Values, and Outlier Report

Overview

This report summarizes exploratory analysis, SQL-based cleaning, missing value handling, and outlier detection for the uploaded ecommerce dataset.

Data quality summary

The dataset has 1200 rows and 14 original columns. CouponCode was the only column with missing values, with 309 missing records. No duplicate OrderID values were found. Date values were complete and TotalPrice matched Quantity multiplied by UnitPrice throughout the dataset.

SQL cleaning and handling

The data was queried in SQL to measure missing values, duplicates, and business summaries. Cleaning included trimming text fields, converting Date to proper datetime format, replacing missing CouponCode entries with NO_COUPON, and creating validation fields such as ExpectedTotal and TotalMismatch.

Outlier detection

Outliers were checked with the IQR rule across Quantity, UnitPrice, ItemsInCart, and TotalPrice. No outliers appeared in Quantity, UnitPrice, or ItemsInCart. TotalPrice had 8 outlier records.

Metric	Value
Rows	1200
Original columns	14
Missing CouponCode	309
Duplicate OrderIDs	0
TotalPrice outliers	8

Top products by revenue

Product	Orders	Avg TotalPrice	Revenue
Chair	178	1098.99	195620.11
Printer	181	1080.73	195612.61
Laptop	173	1110.56	192126.56
Tablet	179	1042.28	186568.95
Monitor	163	1077.62	175651.41

